



ESPE

UNIVERSIDAD DE LAS FUERZAS ARMADAS
INNOVACIÓN PARA LA EXCELENCIA



Homework 4

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Project's Name: Electromechanical Maintenance Management System

Team's Name: Binary Bosses

Problem

In industrial and electromechanical maintenance workshops and companies, operations management is often carried out empirically or through disconnected tools such as spreadsheets and paper records. This lack of a centralized system creates significant operational deficiencies, including a disconnect among clients, technicians, and equipment, which prevents quick linking of machines to clients and efficient assignment of work orders based on technician availability and expertise. Furthermore, technicians experience delayed access to equipment history and technical specifications in the field due to the absence of an immediate identification mechanism.

The inability to automate maintenance scheduling and inventory alerts leads to a predominance of reactive, corrective maintenance over timely preventive inspections, causing unexpected equipment downtime and elevated repair costs. Additionally, historical intervention data is underutilized, missing key opportunities for intelligent failure diagnosis and pattern-based preventive recommendations. Finally, management suffers from a lack of traceability and consolidated reporting to measure productivity and profitability, while clients remain uninformed about real-time equipment status, replaced parts, and service costs.

Overview

The Electromechanical Maintenance Management System is a software solution that helps organise and control maintenance work, in workshops and companies that use equipment. The Electromechanical Maintenance Management System puts information about clients, technicians, equipment, maintenance tasks, parts and work orders into one place.

The Electromechanical Maintenance Management System will help with both corrective maintenance. Preventive maintenance lets technicians plan inspections and services before a failure happens. Corrective maintenance lets technicians fix equipment after a failure is found. The Electromechanical Maintenance Management System also gives equipment history, maintenance schedules, alerts, reports and QR code identification.

The main goal of the Electromechanical Maintenance Management System is to make maintenance tasks more organised, cut equipment downtime, use resources better and give data for technical and administrative decisions.

Background

In industrial and electromechanical environments, equipment reliability is essential for maintaining continuous and safe operations. Machines are exposed to continuous use, wear, and different operating conditions, which can eventually cause failures and affect productivity. For this reason, maintenance is necessary to keep equipment in good working condition and reduce unexpected interruptions.

A key concept related to this problem is preventive maintenance, which involves performing inspections, adjustments, cleaning, and component replacement before a failure occurs. This differs from corrective maintenance, which is performed after a failure has been detected.

Proper maintenance management combines both approaches and uses equipment history to identify recurring problems and plan future interventions.

Maintenance activities also need to follow safety requirements because technicians may work with electrical systems, machinery, tools, and other potential hazards. In Ecuador, the Reglamento de Seguridad y Salud de los Trabajadores y Mejoramiento del Medio Ambiente de Trabajo establishes requirements for protecting workers from occupational risks. This regulation is relevant to the project because maintenance activities should be organized and performed while considering the safety conditions of the workplace.

Therefore, understanding maintenance principles and applicable safety requirements provides the necessary foundation for developing a system that supports preventive planning, records maintenance activities, and helps improve equipment reliability and operational efficiency.